
ME 634: Thermodynamics for Energy Systems

Course Code	ME 634
Course Name	Thermodynamics for Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/M. Tech/MS/PhD
Distribution	Core for M. Tech Energy Engineering (Mechanical), elective for UG/MS/PhD
Approval	10th Senate (Jun 2016)

Course Contents

- **Laws of Thermodynamics**

The first law for open and closed system; steady and transient processes, work and heat transfer; second law of thermodynamics for open and closed systems; Local temperature equilibrium (LTE) Model, entropy maximum and energy minimum principles [10 Lectures]

- **Module II**

Entropy; Concept of reversibility and irreversibility; change in entropy in various thermodynamic processes, entropy balance for closed and open systems, mechanism of entropy generation, entropy generation minimization [7 Lectures]

- **Single and multiphase systems**

Maxwell relations; Clausius-Clapeyron equation; Gibbs-Duhem relation, phase diagrams; phase transition; types of equilibrium and stability; multi-component and multi-phase systems, equations of state. [9 Lectures]

- **Combustion and Thermochemistry**

Stoichiometry of reactions, enthalpy of formation and reaction, adiabatic flame temperature, second law availability analysis of chemical reactions. [6 Lectures]

- **Advanced Thermodynamic Cycles**

Advanced vapour power cycles; advanced gas power cycles, combined cycle power cycles, cogeneration. [6 Lectures]

- **Module VI**

Exergy Analysis of energy systems and case studies. [5 Lectures]

Text and Reference Books

1. A. Bejan, **Advanced Engineering Thermodynamics**, Wiley, 2006.
2. M. J. Moran and H. N. Shapiro, **Fundamentals of Engineering Thermodynamics**, John Wiley and Sons
3. C. Borgnakke, G. Van Wylen and R. E. Sonntag, **Fundamentals of Thermodynamics**, Wiley India